

VidyaSetu — Version 0.1.0

Executive Summary

VidyaSetu is a high-fidelity, multi-tenant **Software as a Service (SaaS)** Education ERP and CRM platform designed to serve multiple coaching institutes and academies. The platform enables centralized platform administration, subscription management, tenant billing, feature gating, and custom-branded tenant portals.

This release represents the **Initial SaaS Frontend Release (Version 0)**. It is a fully client-side Single Page Application (SPA) built with React and Vite, showcasing SaaS multi-tenancy flows, subscription cycles, and role-based tenant portals.

Version 0 at a Glance

This version introduces the entire foundational frontend architecture, including:

- **SaaS Tenant Console:** Centralized workspace for platform owners to register tenants (`TenantsManager.tsx`), monitor branches, and toggle account statuses.
 - **Subscription & Billing Desk:** Configure pricing tiers (`SubscriptionPlans.tsx`), assign plans to tenant instances (`TenantSubscriptions.tsx`), and manage billing cycles.
 - **Role-Based Portals:** Distinct experiences tailored for different users via scoped components:
 - `saas-admin` : Platform owners.
 - `inst-admin` : Institute owners managing their tenant branches.
 - `branch-admin` : Branch managers scoped to specific locations.
 - `counsellor` : Focus on CRM pipelines, admissions, and lead generation.
 - `teacher` : Faculty portal with assignments, attendance, and exam grading.
 - `finance` : Accountants handling expense ledgers and fee configurations.
 - **CRM Lead Pipeline:** Interactive multi-stage admissions funnel (`LeadsAdmissions.tsx`) and student registration (`StudentRegistration.tsx`).
 - **Institute Setup Modules:** Full masters management for Courses (`CourseSetup.tsx`), Branches (`BranchSetup.tsx`), Subjects (`SubjectSetup.tsx`), Classrooms (`ClassroomSetup.tsx`), and Batches (`BatchSetup.tsx`).
 - **Academic Scheduler & Tracking:** Attendance rosters (`Attendance.tsx`), exam marks (`ExamMarks.tsx`), and assignment grading (`Assignments.tsx`).
 - **Finance Management:** Dedicated expense ledger (`ExpenseLedger.tsx`), expense vouchers (`ExpenseVoucher.tsx`), and fee collections (`Fees.tsx` , `FeesMaster.tsx` , `FeeConfigurator.tsx`).
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SaaS Release Highlights

1. **Multi-Tenant Scoping:** Role routing and data displays are dynamically scoped based on the user's logged-in tenant. Platform owners can manage all tenants, while tenant owners and staff can only access their own tenant environment.
2. **Subscription & Plan Gating:** Simulated feature and capacity limits (max branch count, max student intake, allowed modules) determined by the tenant's subscription plan.
3. **Comprehensive Component Architecture:** Granularly separated component structure organized by user role (e.g., `src/components/finance` , `src/components/teacher` , `src/components/branch-admin`) ensuring maintainability and strict access boundaries.

4. **Local API Simulation:** A functional mock JSON backend environment (`server/db.json`) allowing immediate development without a live database.
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Frontend Features & Modules

SaaS Platform Administration

- **Tenants Manager:** Interactive registry to manage, register, and suspend tenant academy accounts (`TenantsManager.tsx`).
- **Plan Master:** Define subscription packages, edit pricing details, and set limits.
- **Tenant Subscriptions Manager:** Assign packages to active tenants, update billing cycles, track renewal indicators, and manage upgrades (`TenantSubscriptions.tsx`).

Role-Based Dashboards & Workspaces

- **Unified Dashboard (`Dashboard.tsx`):** Renders statistics widgets with chart indicators based on user role.
- **Counsellor Panel:** Focuses on CRM pipelines, lead metrics, follow-up schedules, and pending registrations.
- **Finance Panel:** Focuses on cash inflow/outflow counters, expense vouchers, and fee collections.
- **Teacher Panel:** Highlights assignments, attendance taking, exam marks, and active courses.

CRM & Student Management

- **Admissions & Leads:** Dedicated pipeline (`LeadsAdmissions.tsx`) to log enquiries, follow up, and allocate batches.
- **Student Directory:** Complete student registry (`Students.tsx`) with registration stepper (`StudentRegistration.tsx`).

Academic & Institute Setup

- **Structure Management:** Define Branches, Classrooms, Courses, and Subjects.
- **Batch Configuration:** Setup academic batches and assign students and teachers.
- **Staff Directory:** Register and manage faculty and admin staff (`StaffCreate.tsx` , `Users.tsx`).

Finance & Ledger Registry

- **Fees Master:** Configure default course plans, fee structures, and tax parameters (`FeesMaster.tsx`).
 - **Dynamic Fee Configuration:** Specialized component for building custom fee plans (`FeeConfigurator.tsx`).
 - **Expense Tracking:** Track branch cash flows with debit/credit categorization and detailed voucher logs (`ExpenseLedger.tsx` , `ExpenseVoucher.tsx`).
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Technical Implementation

Core Technologies

- **Framework:** React 19.2.7 (Client-Side Rendering)
- **Build Tool:** Vite 8.1.1
- **Routing:** React Router DOM 7.18.1
- **Icons:** Lucide React 1.25.0
- **Styles:** Tailwind CSS 3.4.19 & PostCSS 8.5.20

Development Environment

- **Local Data API:** Runs `json-server` on port `3002` watching `server/db.json` to handle mock database actions.
- **Scripts:** `npm run dev:all` starts both the Vite dev server and the JSON mock API concurrently.